

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `wave 3.1` offset-aware probe campaign into the official `TE Curve Verification Pipeline` offline verification matrix.

Decision

- `wave 3.1` is verified as an exploratory offset-aware sequential residual baseline.
- No `wave 3.1` candidate is promoted over the current accepted `TE Curve Verification Pipeline` leaders.
- The project still maintains three parallel best surfaces: `Fw` , `Bw` , and `global` .
- The strongest paper-derived forward baseline remains `paper_retuned_best_Fw` .
- The strongest paper-derived backward baseline remains `paper_retuned_best_Bw` .
- The strongest repository-owned neural branch remains `wave 2.2` `periodic_gru_sequence_Bw` for backward-only evaluation and `periodic_gru_sequence_global` for bidirectional neural evaluation.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary:
`output/validation_checks/track2_reference_comparison/.../validation_summary.yaml` ;
- per-condition metrics:
`output/validation_checks/track2_reference_comparison/.../per_condition_metrics.csv` ;
- best-model collage report generated by the operator launcher:
`doc/reports/analysis/track2/best_model_collage_report/[2026-06-04]/` ;
- multi-model curve comparison report generated by the operator launcher:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-04]/` .

Candidate Refresh

The refresh added 3 registry-backed wave 3.1 candidates:

Surface	Candidate	Registry
global	sequential_residual_offset_probe_global	output/registries/families/sequential_residual_offset_probe/latest_family_best.yaml
Fw	sequential_residual_offset_probe_Fw	output/registries/families/sequential_residual_offset_probe_fw/latest_family_best.yaml
Bw	sequential_residual_offset_probe_Bw	output/registries/families/sequential_residual_offset_probe_bw/latest_family_best.yaml

The matrix now contains 114 candidates. The incremental operator run used the completed wave 2.3 refresh as its 111 -candidate baseline and evaluated only the 3 new wave 3.1 candidates.

Current Leaders

Scope	Current strongest candidate	MAE [deg]	RMSE [deg]	Mean [%]
forward overall	rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372
backward overall	periodic_gru_sequence_Bw	0.002392	0.002639	5.466
paper-derived forward	paper_retuned_best_Fw	0.001839	0.002041	4.109
paper-derived backward	paper_retuned_best_Bw	0.003675	0.004284	7.572
repository static global	tree_global	0.003144	0.003533	6.854
repository global neural	periodic_gru_sequence_global	0.002704	0.002949	6.139

Wave 3.1 Result

Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]
Fw	sequential_residual_offset_probe_Fw	0.003377	0.003799	7.487
Bw	sequential_residual_offset_probe_Bw	0.003636	0.004065	7.952
global forward slice	sequential_residual_offset_probe_global	0.003425	0.003839	7.599
global backward slice	sequential_residual_offset_probe_global	0.003646	0.004080	7.981
global combined	sequential_residual_offset_probe_global	0.003536	0.003959	7.790

The offset-aware sequential residual probe is competitive with the older Wave 1 neural baselines but does not beat the current curve-following leaders. Its main value is diagnostic: the residual-offset branch did not close the gap to the Wave 2.2 periodic sequence family, so the next offset-aware branch should not simply add a sequential residual head and expect promotion.

Visual Evidence

The operator launcher regenerated the dated collage and overlay bundles for 2026-06-04 . Those report builders still require a follow-up optimization to render dedicated Wave 3.1 visual sections without rerunning an expensive full visual pass. The official decision above therefore uses the refreshed directional matrix as the canonical evidence and treats the visual bundle as a supporting baseline refresh, not as the final Wave 3.1 visual closeout.

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-04	Wave 3.1 offset-aware probe refresh	3 global , Fw , and Bw sequential residual offset candidates	included in the 114 - candidate matrix	dated collage and overlay bundles regenerated; dedicated Wave 3.1 visual sections deferred to optimized visual tooling	verified exploratory baseline; not promoted
2026-05-28	Wave 2.3 residual harmonic temporal hybrid refresh	18 global , Fw , and Bw residual harmonic GRU/LSTM candidates	included in the 111 - candidate matrix	collage and overlay reports refreshed with Wave 2.3 sections	verified exploratory baseline; not promoted over Wave 2.2 or accepted TE Curve Verification Pipeline baselines
2026-05-26	Wave 2.2 harmonic temporal hybrid refresh	periodic temporal convolution, GRU, and LSTM global , Fw , and Bw candidates	included	collage and overlay reports refreshed	strongest repository-owned neural branch
2026-05-24	Wave 2.1 temporal entry refresh	temporal convolution, GRU, and LSTM global , Fw , and Bw candidates	included	visual reports refreshed	verified exploratory baseline

Closeout Decision

Wave 3.1 does not change the accepted TE Curve Verification Pipeline baseline. The best current direction-parallel decision remains:

- Fw : paper-derived rcim_retuned_GBM19_Fw is the strongest overall forward candidate; among repository-owned neural branches, Wave 2.2 periodic sequence models remain the reference family.
- Bw : periodic_gru_sequence_Bw remains the strongest practical backward candidate.
- global : periodic_gru_sequence_global remains the strongest repository-owned bidirectional neural candidate, while tree_global remains the strongest repository-owned static baseline. The next modeling step should move from this single residual-offset probe to a more explicit offset calibration or multi-task offset/shape training branch, with dedicated visual tooling that can render new source groups without rerunning the full legacy collage pipeline.